STAINLESS STEEL

FLANGES

Rout Enterprises

Industries we serve:

- ☑ Pipeline engineering
- ☑ Oil & gas – upstream & downstream
- ☑ Mechanical & plant engineering
- ☑ Ship building
- ☑ Boiler and Pressure Vessels
- ☑ Fertilizer industry
- ☑ Waste water
- ☑ Chemical & petrochemical
- ☑ Power industries (nuclear, natural gas, wind, solar)
- ☑ Aerospace
- ☑ Food processing

Makeability Chart:

Dimensional specifications with size range			
Specification	**Item description**	**Size Range**	**Pressure Class**
ASME – B16.5	Slip-on flange, Welding neck flange, Blind flange, Threaded flange, Lap joint flange, RTJ flange	½" to 24" ½" to 12	150, 300, 400, 600, 900, 1500 2500
ASME – B 16.47	Welding neck flange, Blind flange	26" to 40" (A-MSS) 26" to 30" (A-MSS) 26" to 36" (B-API) 26" to 30" (B-API)	150, 300, 400, 600, 900, 75, 150, 300, 400, 600 900
ASME – B 16.36	Welding neck flange Slip-on flange, Threaded flange	1" to 24" 1" to 12"	300, 400, 600, 900, 1500, 2500
DIN FLANGE	DIN2512, DIN2513, DIN 2514, DIN2527, DIN 2566, DIN2573, DIN2576, DIN2627, DIN2628, DIN2629, DIN2631, DIN2632, DIN2633, DIN2434, DIN2635, DIN2637, DIN2628, DIN2641, DIN2642, DIN2655, DIN2656, DIN2673	10 to 1000	PN6 to PN 320
Pr EN 1092 – 1	Plate flange (01) Loose plate flange (02, 04) / Blind flange (05) Welding neck flange (11)/ Hubbed slip-on flange (12) Hubbed threaded flange (13)/ Integral flange (21), weld-on collar (32) Lapped pipe end (33) Weldneck coller (34) Welding ring neck (35)	10 to 1000	ISO-PN2.5 to PN100

Specification	Item description	Size Range	Pressure Class
EN 1759 – 1	Plate flange (01)/ Blind flange (05) Welding neck flange (11)/ Hubbed slip-on flange (12) Hubbed threaded flange (13)/ Hubbed socket weld (14) Loose Hubbed flange (15) Integral flange (21)	½” to 24” (15 to 600)	150, 300, 400, 600, 900, 1500 2500 (12”)
AFNOR NEF 29-20	Plate flange (01)/Loose plate flange (02, 04)/ 3Blind flange (05) Welding neck flange (11)/ Hubbed Slip-on flange (12) Screwed flange (13) Hubbed socket Welding (14) Loose Hubbed (15)/Integral flange (21)	10 to 600	ISO-PN2.5 to PN420
JIS B2220/B2291	Slip-on plate flange (SOP), Slip-on Hub (SOH), Socket Welding flange (SW), Welding neck flange (WN), Lap joint (LJ), Threaded flange (TR)/ Itegral flange (IT) Blank flange (BL)	10 to 600	5K, 10K, 15K, 20K, 30K
BS4504 SECTION 3.1	Welding neck (111)/Slip-on plate (101)/ Screwed boss (113)/ Slip-on boss (112)/ Blind plate flange (105)	0.375” To 36”	PN2.5 to PN40
BS4504 PART 1	Welding neck (2), Plate slip-on (30), Screwed boss (4), Slip-on boss (5),	0.375” to 36”	PN2.5 to PN 400
BS 1560 SECTION 3.1	Blind plate flange (8) Welding neck (111), Socket weld (114), Screwed boss (113), Slip-on boss (112), Blind flange (105), Lapped flange (115)	½” to 24”	150 LBS to 2500 LBS
BS 10: 1962	Welding neck, Plate Slip-on, Screwed boss, Slip-on boss, Plate blind flange	½” to 36”	TABLE D/E/F/H
SABS/SANS 1123	Integral (1), Welding neck (2), Plate for welding (3), Screwed boss (4), Slip-on boss (5) Plate blank (8)	10 to 1000	250, 600, 1000, 1600, 2500, 4000 kPa
AWWA C228	Ring type Slip-on flanges, Blind flanges	2” to 40”	SA (50Psi), SB (86Psi), SD (150/75Psi), SE (275Psi), SF (300Psi)
Italian standard UNI FLANGES	Plan flanges (2276, 2277, 2278, 6083, 6084) Welding neck (2280, 2281, 2282, 2283, 2284, 2285, 2286) Lapped flanges (6088, 6089, 6090, 2999, 2300) Blind flanges (6091, 6092, 6093, 6094, 6095, 6096, 6097) Threaded flanges (2253, 2254)	15 to 200	PN 6 to PN 40
Norwegian Standard (NS)	Blind flanges, Plain welding flanges	150 to 800	PN 6 to PN 40

Dimensional specifications with size range			
Specification	**Item description**	**Size Range**	**Pressure Class**
Flanges to VSM	Flat flange for welding (18695, 19696,		PN 6 to PN 16
(Swiss) Standards	18697) Weld neck (18716, 18718), Blind flange (18703)		
AS 4087 AS 2129	Blank, Plate, Screwed, weld neck (W.N/W.O.N), Integral flange blank, Plate, Boss, Weld neck, Integral flange	50 to 1000 15 to 1000	PN16, PN21, PN35, TABLE D & E
GOST	Plate flange (GOST 12820-80), Weld neck flange (GOST 12821-80), Blind flange (GOST 12836-67)	10 to 1000	0.6MPa to 6.3 MPa
ASME/ANSI B 16.9 MASS SP-43	STUB END – TYPE A,B,C (Short pattern)	½" to 24"	All Schedules

Grade:

ASTM182/182M ASMESA182/182M ASTM A403/403M ASTM 815/815M	F 304, F304L, F304H, F310, F316, F316L, F316H, F347, F347H, F317, F317L, F321H, F316Ti, F347 , F347H, F51, F60 & F53 S31803, S32205
JIS G3214	SUS F304, SUS F304L, SUS F304H, SUS F316, SUS F316L, SUS F316H, SUS F347, SUS F347H, SUS F317, SUS F317L, SUS F321, SUS F321H
NF E29 – 204 (AFNOR)	BF Z7 CN 18-09 (F304), BF Z3 CN 19-09 (F304L), BF Z7 CND 17.11.02 (F316), BF Z3 CND 17.11.02 (F316L), BF Z6 CNDT 17-12 (316 Ti), BF Z6 CN Nb 18-10 (F347), BF Z6 CNT 18-10 (F321)
DIN 10222-5	1.4301, 1.4306, 1.4307, 1.4541, 1.4571, 1.4401, 1.4404, 1.4410, 1.4438, 1.4450, 1.4462
OTHER PRODUCTS	Long welding neck, Plate flanges (Light weight pattern), Backup flanges (BEV & STD), Beveled ID flanges, Reducing flanges, Case-end

Flanges bore schedules:

S5S, S10, S10S, S20, S30, S40, S40S, S60, S80, S80S, S100, S120, S140, S160, STD, XS, XXS

Flanges facing types:

Flat face, Raised face, Large Raised face, Ring joint, Tongue & Groove face (large & small), Male-Female face

Flanges face finish:

Stock Finish serration, continuous spiral groove (Phonographic) Serration, concentric Serration, Smooth Finish

Manufacturing facilities:

- ☑ Raw material cutting: band saw
- ☑ Open / Close Die Forging & Rolling: Hammer / Press / Ring Rolling
- ☑ Machining: CNC lathes/SPM lathes
- ☑ Bolt Hole drilling: High speed CNC Vertical Machining Center
- ☑ Heat treatment: Solution Annealing (batch annealing furnaces)

Testing facilities:

- ☑ Tensile testing machine (screw driven & Computerized)
- ☑ Impact testing machine (RT & Sub zero)
- ☑ Hardness testing machine
- ☑ Microscope (Image analyzer with color view camera, motorized stage & inclusion inspector)
- ☑ Mobile spectrometer for PMI test : X-ray fluorescence test-gun & OES-Optical Emission Spectrometry
- ☑ Surface roughness tester
- ☑ IGC test facilities
- ☑ Dye penetrate test
- ☑ Ultrasonic testing machine
- ☑ Radiation servo meter
  - Dynamic radiation detection system with weighbridge portal monitors
  - Gamma ray spectroscopy – identifies nuclides & quantifies contamination
  - Portable radiation detection servo meters

Packing:

- ☑ Protect raised face and bevel end with plastic protection caps
- ☑ Using plywood boxes size 800 x 1200 mm (Euro pallet) for shipment

Blind Flange

Weld Neck Flange

Slip-on Flange

Socket Weld Flange

Threaded Flange

Lap Joint Flange

Orifice Flange

Stubend